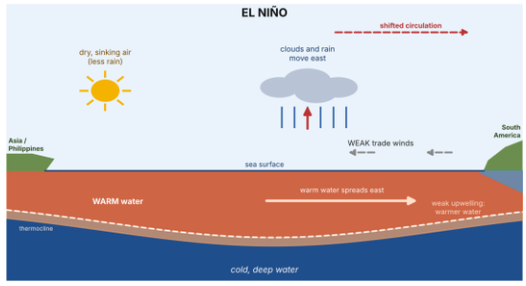
DAILY LESSON PLAN	School	**Taguig National High School**	Grade Level	**10**
	Teacher	**Ian B. Corocoto**	Learning Area	**SCIENCE (Earth Science)**
	Teaching Date and Time	**Day 2** **45 minutes**	Quarter	**Second Term (Week 9)**

I. OBJECTIVES		
A. Content Standards	Climate change and its impacts on the environment and people pose serious challenges which require solutions and action at local and global levels.	
B. Performance Standards	By the end of the term, learners gather information from secondary sources to describe rapid changes that are occurring in local and global climate patterns and propose solutions to address these changes at the local and global levels by drawing on awareness, responsible personal behavior to conserve materials and energy, and through the better societal management of the natural resources of the country.	
C. Most Essential Learning Competencies/ Objectives	At the end of the teaching-learning process, 100% of the students with at least 80% proficiency are expected to: 1. describe how El Niño changes the trade winds, sea-surface temperature and rainfall of the Pacific, and how these changes may impact the weather systems of the Philippines.	
II. CONTENT	**Global Interactions: El Niño – The Warm Phase of ENSO and Its Impact on Philippine Weather**	
III. LEARNING RESOURCES		
A. References	DepEd Grade 10 Science Budget of Work, Second Term, Week 9 – Global Interactions: "Describe how global climatic phenomena, such as the El Niño Southern Oscillation, may impact weather systems." PAGASA (https://www.pagasa.dost.gov.ph) – ENSO Alert and Warning System; El Niño advisories NOAA Climate.gov – "El Niño & La Niña" (https://www.climate.gov/enso) GMA News Online – report on PAGASA's El Niño advisory, 2024 (https://www.gmanetwork.com/news/scitech/weather/899821/pagasa-el-nino-shows-signs-of-weakening-la-nina-watch-to-begin/story/) SunStar Cebu – PAGASA definitions of dry condition, dry spell and drought (https://www.sunstar.com.ph/cebu/pagasa-cebu-under-dry-spellnot-drought) Materials: laptop and projector, "Normal" and "El Niño" diagrams, envelope of 7 chain-reaction cards per group, activity sheet, manila paper, pentel pens	
IV. PROCEDURES	**Teacher's Activity**	**Student's Activity**
	Classroom Conditioning (Let the teacher count 1, 2, 3)	(As the teacher counts, the students will: 1- arrange the chairs, 2- pick up the pieces of paper, and 3- be ready to pray)
	Prayer Let us pray.	(The students will pray.)
	Greetings Good morning, class! Please be seated.	Good morning, Sir! It's nice to see you. Thank you.
	Checking of Attendance Who is absent for today?	The class secretary will report the attendance for today.
A. Reviewing previous lesson or presenting the new lesson (ELICIT) *4 minutes*	Let us review yesterday's lesson through **"QUICK FIRE."** I will ask a question and you will answer in one breath. 1. In normal conditions, which way do the trade winds blow?	From east to west.
	2. Where is the warm pool located?	In the western Pacific, near the Philippines and Indonesia.
	3. What do we call the rise of cold, nutrient-rich water near South America?	Upwelling.
	4. What does ENSO stand for?	El Niño Southern Oscillation.

	Very good! For your assignment, who would like to share what an elder or family member told you about an unusually dry or rainy year?	(Two or three students share.) For example: "My lola said that in one very dry year our well dried up."
	Thank you! Yesterday's last question asked: "If the trade winds became very weak, what would happen to the pile of warm water in the west?" Today we will find out in detail.	It would spread back toward the east.
B. Establishing a purpose for the lesson (ENGAGE) *4 minutes*	**"NEWS FLASH!"** Listen to this report based on news about the 2023–2024 El Niño: ☐ PAGASA warned that about **40 provinces** could experience drought, with March and April as the peak months. ☐ As of March 2024, agricultural damage was reported at about **₱1.24 billion**, affecting more than **29,000 farmers and fishermen** and more than **26,000 hectares** of crops. ☐ A town in Oriental Mindoro declared a **state of calamity** because of severe water shortage. What do you think caused so many places to be dry at the same time?	Maybe there was very little rain for many months. It might be El Niño, Sir. (Answers may vary)
	Good! Notice that the problem was **not just one town**. It affected many provinces at the same time. That is a clue that the cause is **bigger than local weather**: a global climatic phenomenon. Our lesson objective is to: **describe how El Niño changes the trade winds, sea-surface temperature and rainfall of the Pacific, and how these changes may impact the weather systems of the Philippines.**	(The students read the objective.)
C. Presenting examples/ instances of the new lesson (EXPLORE) *10 minutes*	Let us become "climate detectives." **ACTIVITY: "EL NIÑO CHAIN REACTION"** (Group of 5; 8 minutes; one envelope per group) **Objective:** To arrange the events of El Niño in the correct order. **Mechanics:** 1. Each group will receive an envelope with 7 cards. Each card is one event of El Niño. 2. Arrange the cards in the order in which the events happen. Use arrows on the manila paper. 3. Paste the cards in order and add one real-life example that shows the last event. 4. Choose a reporter to explain your chain. (The teacher distributes the materials. See the attached activity sheet for the cards.)	(The students arrange the cards.)
	(The teacher walks around and gives hints if needed.) Hint: "What is the first thing that changes in the Pacific when El Niño starts?" Hint: "Where does the rain go when the warm water moves?"	The trade winds weaken first, so the warm water can move east.
	Time is up! Group reporters, please present your chain. (Two groups report; the class checks the others.)	(The reporters present their chains.)
D. Discussing new concepts and practicing new skills #1 (EXPLAIN)	Let us check your chains against the diagram of El Niño.	The trade winds become weak.

10 minutes	 Compare this with yesterday's normal conditions. **What is the first change?**	
	Correct! This is the "trigger." When the trade winds weaken, what happens to the warm water that was piled up in the west?	It flows back toward the central and eastern Pacific.
	Yes! And what happens to the upwelling of cold water near South America?	It weakens, so the sea surface there becomes warmer than normal.
	Excellent! Scientists declare El Niño when the sea-surface temperature in a key area of the central-eastern Pacific stays about **0.5 °C or more above average** for several months. Where do you think the rising air and heavy rain move?	Toward the center and the east of the Pacific, following the warm water.
	Right! The rainy area moves away from the western Pacific. Over the Philippines, the air tends to sink more, so fewer rain clouds form. How would you describe the rainfall over the Philippines during El Niño?	Below normal; it becomes drier.
	Very good! PAGASA uses terms to describe how dry it gets. (The teacher shows the table.) • **Dry condition** – 2 consecutive months of below-normal rainfall (21–60% below average) • **Dry spell** – 3 consecutive months of below-normal rainfall, or 2 consecutive months of way-below-normal rainfall (more than 60% below average) • **Drought** – 5 consecutive months of below-normal rainfall, or 3 consecutive months of way-below-normal rainfall	(The students copy the definitions.) Sir, so drought is the most serious one.
	Exactly! Aside from less rain, El Niño may also bring: • **higher temperatures and heat index**, • **lower dam levels** and water rationing, • **damage to crops** such as rice and corn, • **less hydropower** from dams, • **coral bleaching** and changes in fish catch because of warmer seas, • **tropical cyclones** forming farther east, so fewer may reach the Philippines in some seasons, but those that form can become strong.	(The students take notes.)
	Important note: not every dry period is caused by El Niño, and every El Niño event is different. Its effects vary by season and by region. That is why PAGASA issues **El Niño Watch, Alert,** and **Advisory** statements to warn the public.	So El Niño is only one of the causes of dry weather.
E. Developing mastery (leads to Formative Assessment) (EXPLAIN) *3 minutes*	Why do you think El Niño affects the Philippines more strongly than places in the middle of the continent?	Because the Philippines is on the western edge of the Pacific, near the warm pool, so when the rain area moves away, we feel it.
	Excellent reasoning! Metro Manila gets most of its water from Angat Dam in Bulacan. What could happen to the water supply if there is little rain for several months?	The dam level could go down and there may be water rationing.
F. Finding practical application of concepts and skills in daily living (ELABORATE) *4 minutes*	**"IMPACT WEB"** (Think-Pair-Share) In the center write **EL NIÑO**. Around it, write one effect on each of: (a) water, (b) food, (c) health, (d) school or work. Then, with your seatmate, add one action a family can do.	(a) Water: dams go down; (b) Food: rice and corn dry up; (c) Health: heatstroke; (d) School: classes suspended because of heat. Action: save water, store water, use hats and drink water, plant drought-tolerant

	(The students complete the web on the activity sheet in 3 minutes.)	crops.
	Why should farmers and local governments pay attention to PAGASA's El Niño advisories?	So they can prepare: change planting schedules, store water, and plan assistance.
G. Making generalizations and abstractions about the lesson (ELABORATE) *2 minutes*	Let us summarize with **"3-2-1"**: • Name **3** things that change during El Niño. • Name **2** effects on the Philippines. • Write **1** question you still have.	3 changes: weaker trade winds, warmer central/eastern Pacific, rain moving east. 2 effects: below-normal rainfall and higher temperatures. 1 question: Is there an opposite of El Niño?
	Good question! There is. That will be tomorrow's lesson.	(The students listen.)
H. Evaluating learning (EVALUATE) *5 minutes*	**MULTIPLE CHOICE:** Write the letter of the correct answer on a ¼ sheet of paper. 1. During El Niño, the trade winds are: a) stronger than normal b) weaker than normal c) blowing west to east only d) absent in the Pacific 2. Where does the warm water spread during El Niño? a) Toward the central and eastern Pacific b) Toward the Arctic c) Only to the Atlantic d) Nowhere; it stays the same 3. Which is the most likely effect on the Philippines during El Niño? a) Rainfall above normal in all areas b) Rainfall below normal in many areas c) No change in temperature d) More snow in the mountains	1. B 2. A 3. B
	4. A province experienced 3 consecutive months of below-normal rainfall. How does PAGASA describe this? a) Dry condition b) Dry spell c) Normal condition d) Flood 5. A farmer says: "It is already the rainy season on the calendar, but my rice field is cracking." Which explanation best uses what you learned? a) The sun moved closer to the Philippines. b) During El Niño the rain area shifts eastward, so less rain falls over the Philippines. c) The farmer planted too early. d) Trade winds are stronger, so all the rain stays in the sea.	4. B 5. B
I. Additional activities for application or remediation (EXTEND) *1 minute*	**Assignment:** 1. Choose one: the 2015–2016 El Niño or the 2023–2024 El Niño. Using a reliable news source (PAGASA, DOST, DA, or a national newspaper), list 3 effects on the Philippines. Write the title and date of your source. 2. **Predict:** If El Niño is a weakening of the trade winds, what do you think happens in the Pacific when the trade winds become **stronger** than normal? Draw your prediction. Thank you, class! Goodbye.	(The students write the assignment in their notebooks.) Goodbye and thank you, Sir!
J. REFLECTION		

A. No. of learners who earned at least 80% in the evaluation	
B. No. of learners who require additional activities for remediation	
C. Did the remedial lessons work? No. of learners who have caught up with the lesson	
D. No. of learners who continue to require remediation	
E. Which of my teaching strategies worked well? Why did these work?	
F. What difficulties did I encounter which my principal or supervisor can help me solve?	
G. What innovation or localized materials did I use/discover which I wish to share with other teachers?	

Prepared by:

IAN B. COROCOTO
Student Teacher

Checked:

[COOPERATING TEACHER]
Critic Teacher
Master Teacher 1

Noted:

[HEAD TEACHER]
Head Teacher III

Group: _____ Score: _____

"EL NIÑO CHAIN REACTION" – Cards (cut along the lines and place in an envelope)

Card 1. Over the Pacific, the trade winds weaken.	**Card 2.** Warm water that had piled up in the western Pacific flows back toward the east.
Card 3. The central and eastern Pacific becomes warmer than normal; upwelling of cold water weakens.	**Card 4.** The area of rising air, clouds and heavy rain shifts eastward, away from the Philippines and Indonesia.
Card 5. Over the Philippines, air sinks more and fewer rain clouds form.	**Card 6.** Rainfall stays below normal for several months.
Card 7. Dry spells or drought, water shortage, crop damage and very hot days affect communities.	

Teacher's key (correct order): 1 → 2 → 3 → 4 → 5 → 6 → 7. (Cut the cards apart and shuffle before putting in envelopes.)

"IMPACT WEB" – Graphic Organizer

Area	Effect of El Niño (in the Philippines)	What can a family or community do?
Water		
Food		
Health		
School / work		

Rubric for the Chain Reaction Activity (20 points)

Criteria	4	3	2	1
Correct sequence	All 7 cards correct	5–6 correct	3–4 correct	0–2 correct
Arrows and neatness	Clear and neat	Mostly clear	Unclear	Missing
Real-life example	Accurate and specific	Accurate but general	Partly accurate	None
Reporting	Clear, confident, accurate	Mostly clear	Unclear	Did not report
Cooperation	Everyone participated	Most participated	Few participated	One member only

Answer Key – Day 2 Quiz

1. B 2. A 3. B 4. B 5. B